NIDS MODULE WALKTHROUGH

Measuring Internet Disruptions with the UCSD Network Telescope

From darknet traffic to a national blackout — how the assignment's four tasks fit together

Assignment: nids-ucsdnt-expanse · Tasks 1–4 · Prototype notebook → Slurm batch job → analysis notebook

BACKGROUND

Recent geopolitical tension disrupted the Internet connectivity


BBC

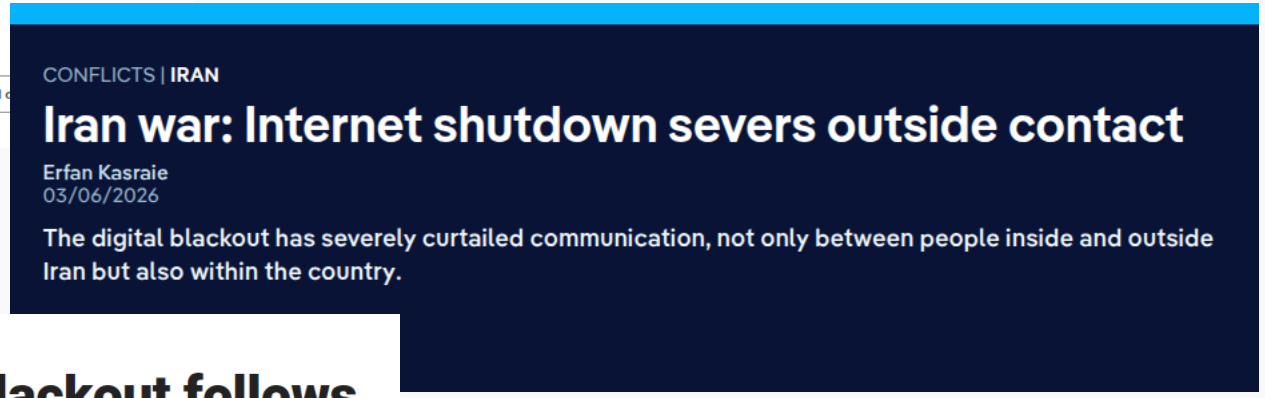
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Internet starts coming back in Iran after months-long blackout

26 May 2026

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Maia Davies and Ghoncheh Habibiazad, Senior reporter, BBC Persian


CONFLICTS | IRAN

Iran war: Internet shutdown severs outside contact

Erfan Kasraie
03/06/2026

The digital blackout has severely curtailed communication, not only between people inside and outside Iran but also within the country.


WAR WITH IRAN

Iran goes dark amid 'regime paranoia,' blackout follows Israeli, US strikes on compound

NetBlocks CEO says connectivity 'flatlined' as regime tried to avoid leaking locations of senior figures

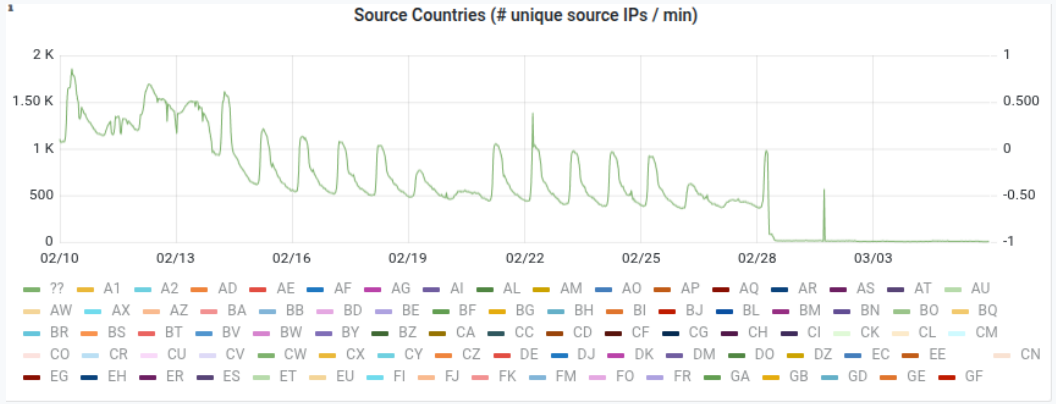
By Emma Bussey · Fox News

Published February 28, 2026 9:31pm EST

BACKGROUND

Why the network telescope can be used to identify outage?

- Sources of unsolicited traffic
  - Scanning
  - DDoS Backscatter
  - …
- These unsolicited traffic also blocked from sending to the Internet during blackouts.



OVERVIEW

What This Walkthrough Covers

1

Network telescope data in FlowTuple format

What a darknet actually sees, and why the Avro file format forces a different Spark pattern than you've used before.

2

The four tasks, one pipeline

Prototype → batch job → analysis — the same per-day logic reused at three different scales.

3

Questions about Iran’s Internet connectivity

What each one is really asking you to notice in the data, not just which cell to fill in.

BACKGROUND

One File Every Five Minutes

FlowTuple records summarize telescope traffic; each field below is used somewhere in the pipeline.

field	meaning
`time`	start of the 5-minute collection window (UTC)
`src_ip`	source IP, packed as a 32-bit integer
`packet_cnt`	packets observed for this record
`prefix2asn`	origin ASN of the prefix covering src_ip
`netacq_country`	geolocated source country

AVRO (FlowTuple dataset)

No column or predicate pushdown — a whole file must be decoded before a country filter can drop non-matching rows.

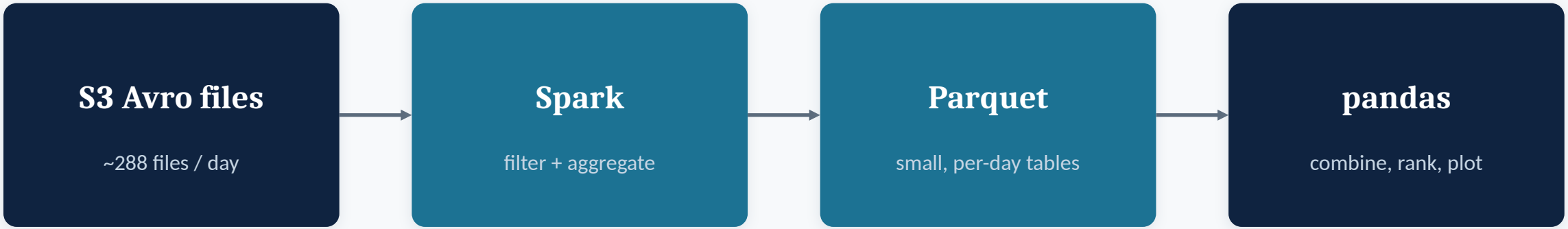
PARQUET (this module's own outputs)

Supports column/predicate pushdown — which is exactly why analysis.ipynb can read the results with plain pandas, no Spark needed.

This is why loading a day's ~288 five-minute files is expensive even when the IR-geolocated subset is small.

BIG PICTURE

Four Tasks, One Pipeline



Tasks 1 & 2
(prototype notebook)

Task 3
(Slurm batch job)

Task 4
(analysis notebook)

Heavy lifting happens once, in Spark. Everything downstream of Parquet is a small table.

DESIGN PATTERN

Two Test Days, Then Twenty-Four

Prototype (interactive)

- 2 test days **(full day)**
- One long-lived Jupyter kernel
- Goal: work out and sanity-check the per-day logic
- Tips: During development phase, use **jupyter-shared-spark** for shorter queue time
  Use **jupyter-shared-spark-large** for generating the results

Batch (Slurm array)

- ~24-day period **(one hour per day)**
- One array task per day, several running concurrently
- Goal: run the proven logic unattended, at scale
- Tips: You can reduce the number of days for testing. The resource requested should be sufficient

Task 3 ports Task 1's logic almost verbatim — the aggregation itself doesn't change, only how and how often it runs.

TASK 1

Filter Once, Compute Four Things

Filter each day's Spark DataFrame to IR-geolocated rows a single time, then reuse it for all four outputs below.

1

Daily summary

total packets, unique source IPs, unique ASNs

2

Source / ASN pairs

distinct (src_ip, prefix2asn) combinations

3

Packets per ASN

who contributes the most traffic that day

4

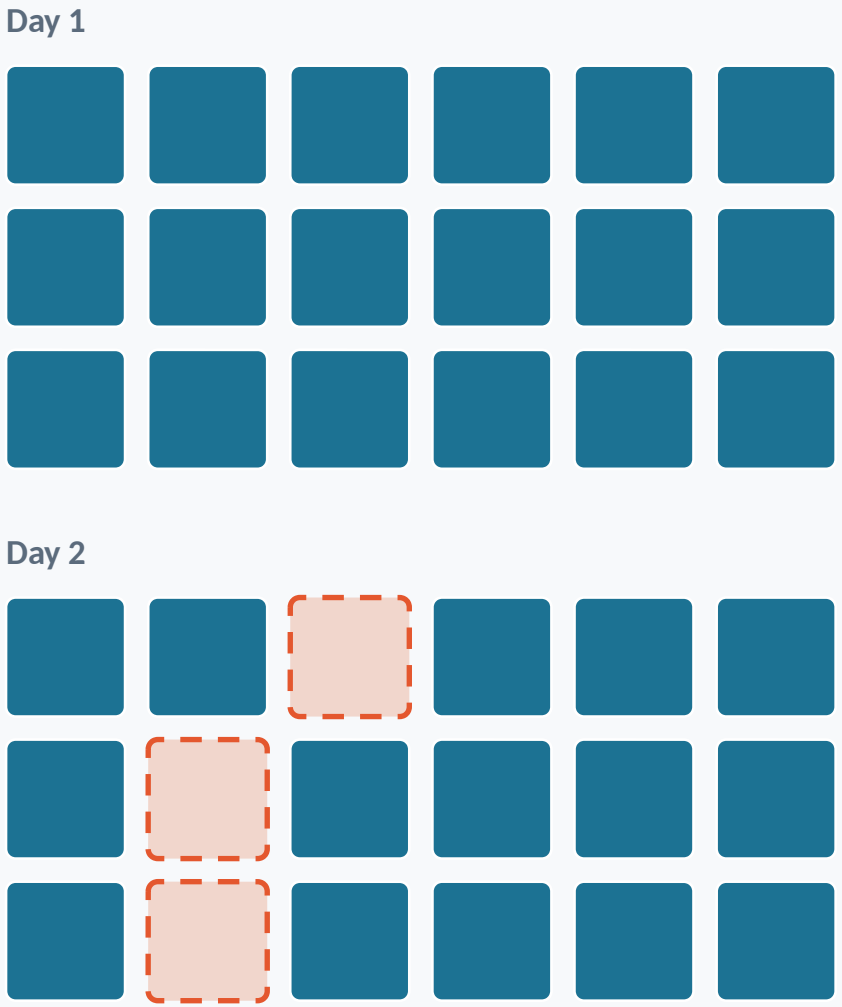
Packets per hour

an hourly trend across the day

TASK 2

Why /24 Subnets, Not IPs or ASNs

- Individual IPs get dynamically reassigned by ISPs day to day — too noisy to mean “disappeared.”
- Whole ASNs are too coarse to see localized changes within them.
- A /24 (256 addresses) is the middle ground: stable enough to track, granular enough to matter.
- Task 2 asks: which (subnet, ASN) pairs were present yesterday but absent today — a set difference across days.

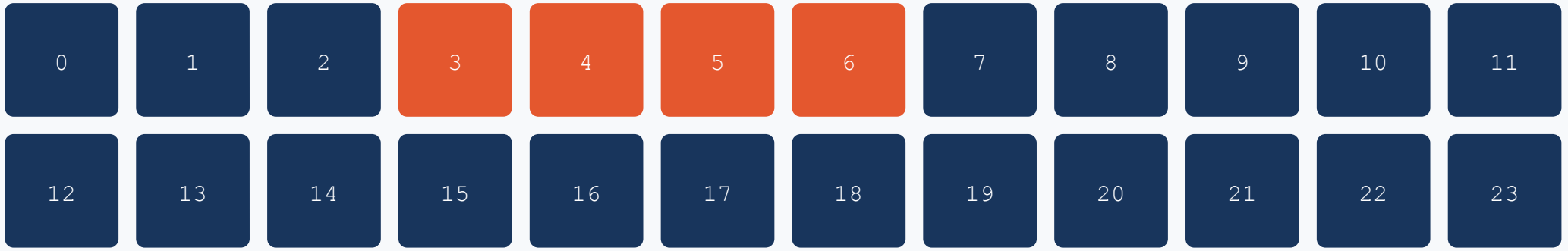


dashed = /24 that disappeared

TASK 3

24 Days, Running at Once — Not One at a Time

- The same per-day logic from Task 1, ported into a standalone script that takes one --date
- A Slurm job **array** runs one task per day — several execute concurrently instead of queuing behind each other
- Each task deliberately samples only the 00:00 UTC hour, to keep the full period's total data volume manageable



highlighted = 4 tasks running concurrently (--array=0-23%4)

TASK 4

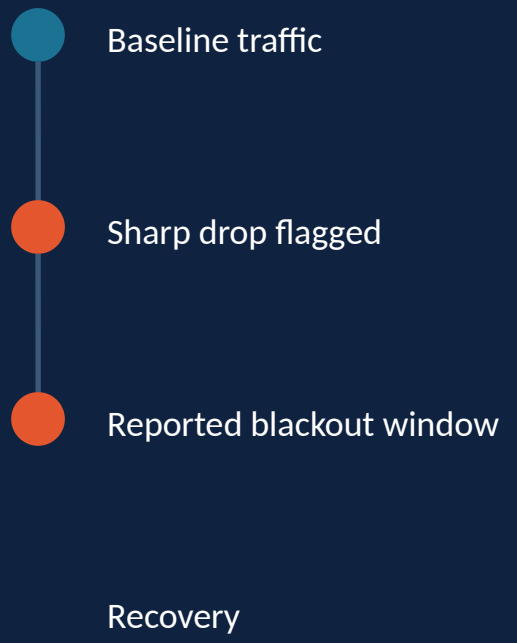
Analysis notebook — Just pandas

- Combine the per-day Parquet files from the batch job into one full-period table
- Rank ASNs by total packets, and separately by distinct /24 footprint
- Flag a day as a “drop” when its traffic falls sharply versus the previous day (pct_change)
- To better visualize the trend, **apply log scale on the y-axis**
- **jupyter-shared-spark** is sufficient to run the analysis

REAL-WORLD ANCHOR

A Drop Alone Isn’t Proof of an Outage

- Traffic can fall for mundane reasons too — a collection gap, a routing change, even a holiday
- The 2026 Internet blackout in Iran article gives an independent, second signal: reported dates and described severity/scope
- Check whether the drop hits most ASNs at once (broad, consistent with a national event) or a concentrated few (targeted, consistent with something else)



Q6 asks you to make exactly this comparison, by hand, against the article.

QUESTIONS

Every Question Ties Back to One Task

Q1	Task 1	How stable is day-to-day “background” telescope traffic?
Q2	Task 2	Why churn at /24 granularity — and what besides an outage explains it?
Q3	Task 3	How does a job array speed things up versus one interactive loop?
Q4	Task 4	Do the top ASNs by packets match the top ASNs by /24 footprint?
Q5	Task 4	Is there a diurnal pattern, and does it line up with local time?
Q6	Task 4	Does your flagged drop match the reported blackout’s dates and severity?

WORKFLOW

What You'll Submit

1. Get Expanse access, clone the repo, launch an interactive Jupyter session
2. Complete ir_traffic_prototype.ipynb — Tasks 1 & 2, against the two test days
3. Port that logic into ir_traffic_batch.py and submit run_ir_traffic.slurm — Task 3
4. Once the array finishes, complete analysis.ipynb — Task 4, full-period view
5. Answer all six questions in the notebooks' markdown cells
6. Commit and push your notebooks and script to GitHub

BEFORE YOU START

Troubleshooting

Size driver memory deliberately

The memory parameter in ('spark.driver.memory', memory) should be smaller than your allocation.
The default in the prototype notebook set it to be 24G (for **jupyter-shared-spark-large**).
Set it to 12g if you use **jupyter-shared-spark.**
Otherwise, you will run into "Connection refused" errors.

Most errors can be recovered by restarting the kernel

"SparkContext was shut down" or "Connection refused" means the OS killed the JVM. You can rerun it by restarting the kernel (after changed configurations). You *do not need to 'scancel' the job*.
S3 Authentication error can be recovered by simply rerunning the cell.

You may not need to re-run the whole code if you obtain some correct parque files

Some jobs might fail in job array. You can re-run those particular day(s) if other days produced correct parque output.

WRAP-UP

Where to Go From Here

- Start with Introduction.md and Datasets.md before opening the notebook
- Each Task-N-….md guide has the exact Spark/pandas pattern you need, spelled out
- The six questions live in the markdown cells right where you'll answer them

README · Introduction · Datasets · PySpark & Parquet · Tasks · Task 1–4 guides